

Case Study: Netflix

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Introduction

Netflix is one of the leading digital streaming platforms in the world with millions of subscribers across many countries. Unlike traditional media companies, Netflix relies heavily on data to operate its platform. Every action on Netflix such as searching for a movie, clicking on a title, pausing a video or watching a show, generates a large amount of user data. This data helps Netflix understand viewer behavior and preferences more accurately. It is also widely known that Netflix operates its streaming infrastructure using cloud services from Amazon Web Services (AWS) which enables the platform to process and manage large volumes of data efficiently.

Big data analytics plays an important role in Netflix's business strategy. By analyzing viewing history and content information, Netflix can identify popular trends and predict what audiences may want to watch. These insights allow Netflix to provide personalized recommendations and improve the overall user experience. Big data also supports marketing and content decisions. Netflix uses data to promote relevant content to users and recommend movies or shows that match their interests. In addition, data analysis helps Netflix decide which original content to produce, reducing the risks associated with content investment and improving customer retention.

The purpose of this report is to examine how Netflix uses big data technologies and analytics to support its recommendation system and business operations. The report discusses the types of data generated by Netflix, the design of its big data system architecture and the challenges related to the five characteristics of big data which are Volume, Velocity, Variety, Veracity and Value. It also explores machine learning techniques used in recommendation systems and explains how scalability, fault tolerance and data security can be maintained in a large-scale streaming platform.

The report is divided into five sections. Section 1 covers the types of data generated by Netflix and their role in personalization. Section 2 outlines the high-level architecture of the big data system. Section 3 addresses how Netflix manages the five characteristics of big data. Section 4 examines machine learning techniques in recommendation systems. Section 5 discusses strategies for scalability, fault tolerance and data security in large-scale streaming systems.

Data Types and Support for Personalization

Netflix generates a massive volume of data that can be categorized by both structured and unstructured data types

Structured Data

Structured data is organized and the format is usually defined before the data is stored. Below are some examples of the structured data that Netflix stores and how it aids personalization.

1. **Explicit Feedback Ratings:** Users can rate titles
 - a. **Support for Personalization:** Netflix uses a *Recommendation System*. There are two types of recommendation systems: collaborative filtering models and content-based filtering (AlgoDaily, 2026).

Collaborative Filtering: Recommendations of other members with similar tastes and preferences on the service (Netflix, n.d.). If user A likes “Breaking Bad” and user B likes “Breaking Bad” and “Gotham”, then “Gotham” is recommended to user A.

Content- Based Filtering: Recommendations of how you rated a title (Netflix, n.d.). The algorithm would find other similar titles based on the title rating you provided.

2. **Viewing Metrics:** Analyzing the user’s viewing history (Netflix, n.d.). Netflix is able to capture how long the user spent on a show, if the user completed the show, or if the user abandoned the show entirely.
 - a. **Support for Personalization:** If the user does not complete watching a show, the system learns to rank it lower for similar profiles as the user

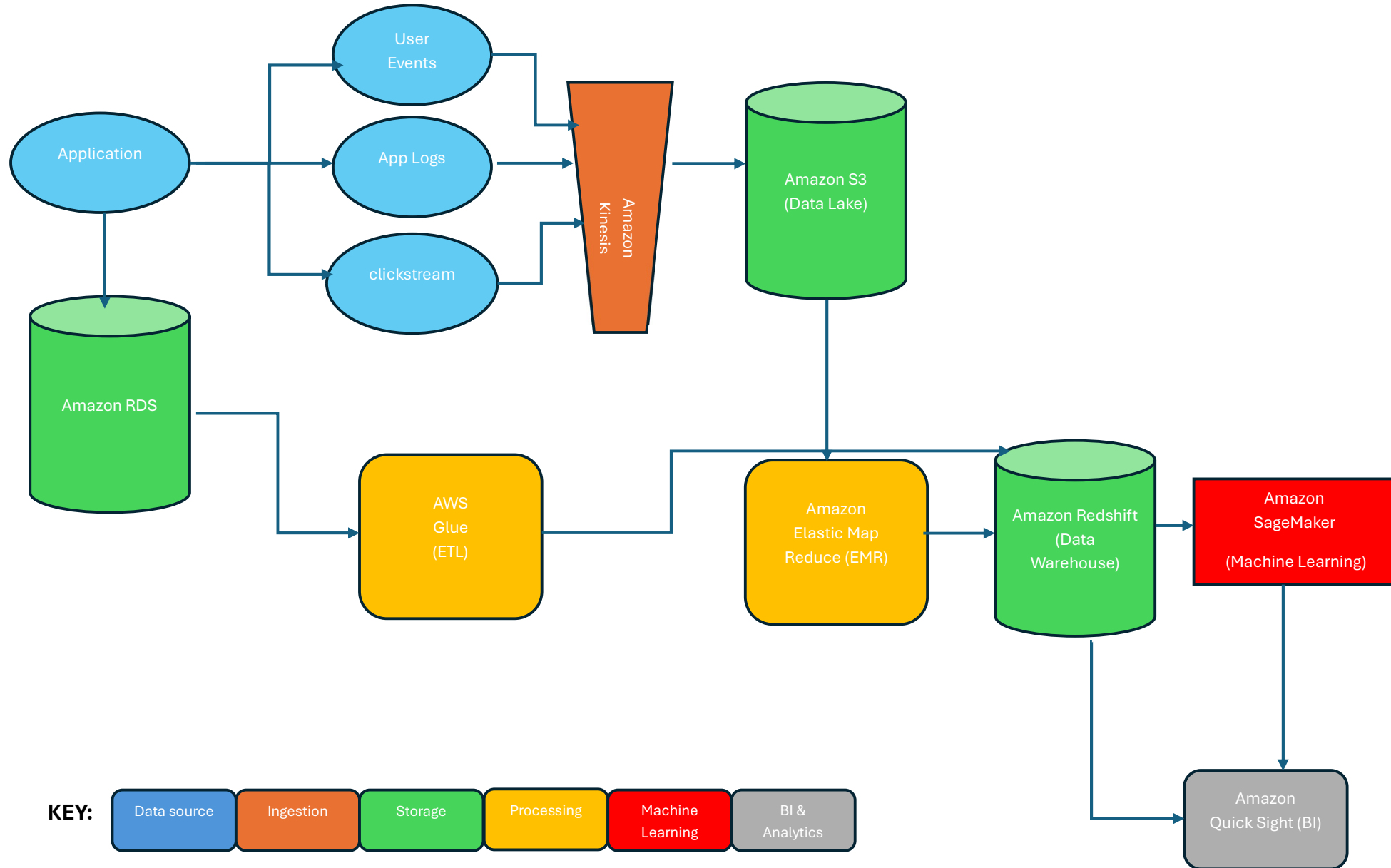
Unstructured Data

Unstructured data is not stored in any predefined format. Below are some examples of the unstructured data that Netflix stores and how it aids personalization.

1. **Search Queries:** The raw text typed in search bar
 - a. **Support for Personalization:** Natural Language Processing (NLP) analyzes the text and understand user’s intent and interests even when the specific title isn’t available and allows the system to suggest related alternatives

2. **Visual Content (Thumbnails):** Netflix stores millions of thumbnails and frames and customizes the thumbnail for every user.
- **Support for Personalization** Netflix uses *Contextual Bandits* (a machine learning algorithm) to match artwork to your specific taste. If a user frequently watches romances, the system might show a thumbnail of a show's lead couple. If they prefer action, it might show an explosion from the same show. (Chandrashekar et al. (2017))

Netflix System Architecture Diagram



Architectural Design Decisions

1. **Amazon Kinesis (Ingestion Layer):** Used for high-speed streaming (Amazon Web Services, n.d.). Captures the data from various sources: logs, clickstreams, user events
2. **Amazon RDS (OLTP Storage Layer):** Amazon RDS is a managed service designed to simplify the setup and scaling of relational databases in the cloud, helping Netflix optimize their total cost of ownership (Amazon Web Services, n.d.).
3. **Amazon S3 (Data Lake- Primary Storage Layer):** This is where everything arrives first. Amazon S3 The central repository or Data Lake that stores massive volumes of raw logs and video assets (Amazon Web Services, n.d.).
4. **Amazon Elastic Map Reduce (Processing Layer):** Used to run data processing and transformation (Amazon Web Services, n.d.). The EMR transforms the data (from Amazon S3) and sends it to Amazon Redshift.
5. **AWS Glue:** ETL process to transform the Netflix data (e.g. calculating monthly totals) and sends it over to the data warehouse (Amazon Redshift).
6. **Amazon Redshift (OLAP Storage Layer):** The Data Warehouse is where structured data is stored for complex SQL Analytics and Business Intelligence. This would be ideal for calculating Netflix style metrics “What is the average watch time per genre”. Amazon Redshift will be used as their OLAP. Amazon Sage Maker will run off of Redshift.
7. **Amazon SageMaker (Machine Learning):** A service that allows data scientists and developers to quickly build, train, and deploy machine learning (ML) models at any scale (Amazon Web Services, n.d.). Netflix will build specialized style algorithms such as a specialized recommendation engine and perhaps even a user churn predictor. Sage maker is also better suited for heavy data work that data scientists require. Ideally, we will build models in Sage Maker and use Amazon Quick Sight for visualization.

8. **Amazon Quick Sight (BI & Analytics):** This will be used for our Business Intelligence (Amazon Web Services, n.d.). Here we will create interactive dashboards and visualizations that enable stakeholders to explore data and gain actionable insights.

Addressing the Challenges of the 5Vs

Machine Learning Technique

What machine learning techniques could be used in Netflix's recommendation system?

@Daniel ML "Contextual Bandits" see Q1-> unstructured -> #2

Explain their role in system design (we use SageMaker for ML)

<https://aws.amazon.com/sagemaker/ai/>

<https://youtu.be/1bt11sLo5Pw>

Ensuring Scalability, Fault Tolerance, Data Security

Our architecture is designed not just as a collection of tools, but as a cohesive system where each component choice directly impacts the platform's reliability and performance. Below is a detailed analysis of how our design meets enterprise-level requirements.

Individual Component Analysis

This table breaks down how each AWS service contributes to our three core pillars: Scalability, Fault Tolerance, and Data Protection.

Feature	Scalability	Fault Tolerance	Data Protection
Amazon Kinesis	Uses shards to scale throughput; operates in on-demand mode for automatic scaling.	Data is replicated across 3 Availability Zones (AZs) automatically to prevent loss	Supports server-side encryption and private VPC endpoints.
Amazon S3	Offers unlimited storage capacity; scales automatically to handle high request rates.	Designed for 99.999999999% durability by storing data across multiple locations.	Supports versioning, MFA delete, and AES-256 (KMS) encryption.
Amazon RDS	Scales horizontally using 3-5 Read Replicas to offload "Read" traffic from the primary DB.	Uses Multi-AZ deployments for synchronous replication and automatic failover in <60s.	Supports versioning, MFA delete, and AES-256 (KMS) encryption.
Amazon EMR	Can resize clusters dynamically based on the workload (e.g., peak processing hours).	If a node fails, EMR automatically replaces it without losing data progress.	Supports Kerberos authentication and encryption for distributed data.

Architectural Level Analysis

Beyond individual tools, the synergy between these components provides high-level system resilience:

- Decoupling with Buffers:** By using **Kinesis** and **S3** as staging areas (buffers), the system becomes highly resilient. For instance, if the **EMR** cluster fails during processing, the raw clickstream data remains safe at the source. This decoupling

ensures that a failure in downstream processing does not lead to permanent data loss (Amazon Web Services [AWS], 2024).

- **S3 as the Single Source of Truth:** Our architecture treats **S3** as a persistent data lake. This allows for clear **Data Lineage**. If a transformation in the database fails or if business logic needs updating, we can re-process the raw data from S3 without affecting the production environment (Vajgel & Murthy, 2022).
- **Integrated Security and Failover:** By combining **RDS Multi-AZ** and **Encryption at Rest (KMS)**, we ensure that the system is not only always online but also compliant with data privacy regulations like GDPR. Even during a complete data center outage, the automated failover mechanisms guarantee continuity with zero data loss (Hoffman, 2023).

Conclusion

Summarize key insights:

- Importance of big data for your case-study
- Role of AI and analytics in business decisions
- Benefits of scalable and secure architecture

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